

Daily Progress Report

Week 5_Day3: The React Lifecycle & useEffect

Program	3-Month MERN Stack Internship Program
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Week	Week_5 (Day 3)
Date	23 July, 2026

Day 3: The React Lifecycle & useEffect

1.1 Topics Studied

- Understanding the React Component Lifecycle (Mount, Update, and Unmount)
- Introduction to the useEffect hook for managing side effects in functional components
- Executing side effects after component rendering
- Using dependency arrays ([]) to control when effects are executed
- Fetching data asynchronously from external APIs using the Fetch API
- Managing asynchronous operations with async/await
- Displaying loading states while asynchronous data is being fetched
- Implementing graceful error handling using try...catch
- Conditional rendering in React based on loading, success, and error states
- Organizing asynchronous logic inside React components following best practices

1.2 Task Description

Developed a React component that retrieves mock post data from a public API immediately after the component mounts. The component utilizes the useEffect hook to perform the asynchronous API request while maintaining separate state variables for fetched data, loading status, and error handling. During the data retrieval process, a loading spinner is displayed to provide visual feedback to the user. If the API request fails, an appropriate error message is presented instead of the content. Upon successful completion, the fetched posts are rendered dynamically in a responsive card-based layout.

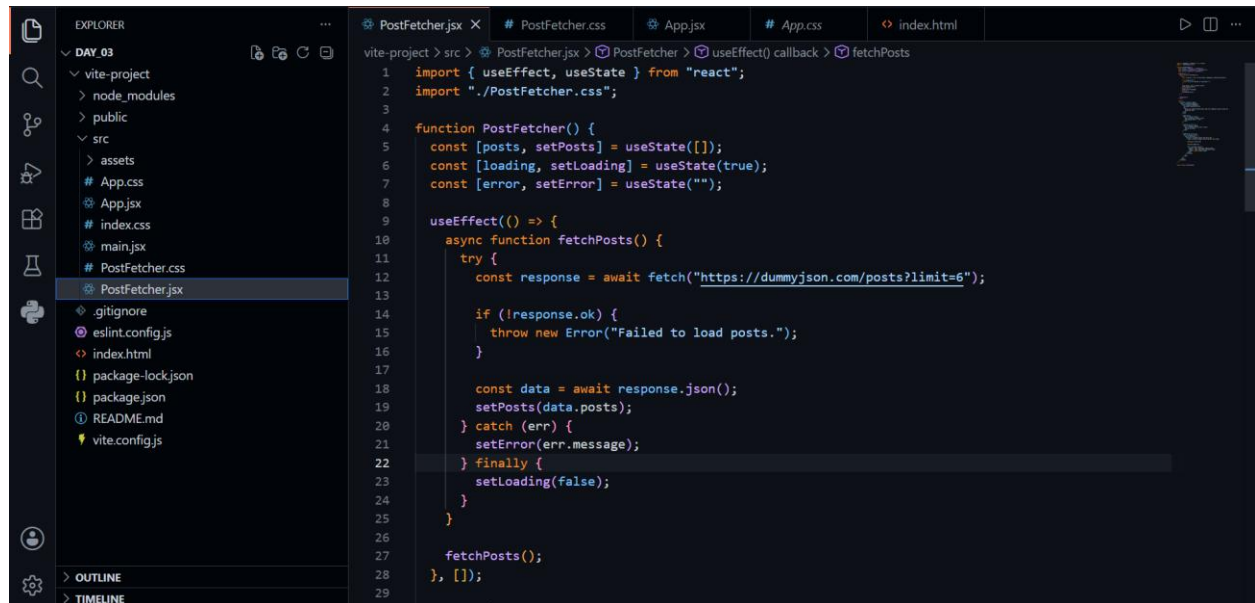
1.3 Implementation Details

- Configured a dedicated Posts component to fetch and display mock data from the DummyJSON API.
- Utilized the useState hook to manage the posts, loading, and error states within the component.
- Implemented the useEffect hook with an empty dependency array ([]) to execute the API request only once when the component mounted.
- Performed the asynchronous API request using the Fetch API with async/await and validated the server response before processing the returned data.
- Implemented try...catch...finally logic to handle API errors gracefully and ensure the loading state was updated after the request completed.
- Applied conditional rendering to display a loading spinner during data fetching, an error message if the request failed, and the fetched posts after a successful response.
- Rendered the retrieved posts dynamically using the .map() method while assigning unique key props to each post for efficient rendering.
- Integrated custom CSS to create a clean, responsive card-based layout with hover effects, loading animations, and user-friendly error indicators.

1.4 Outcome

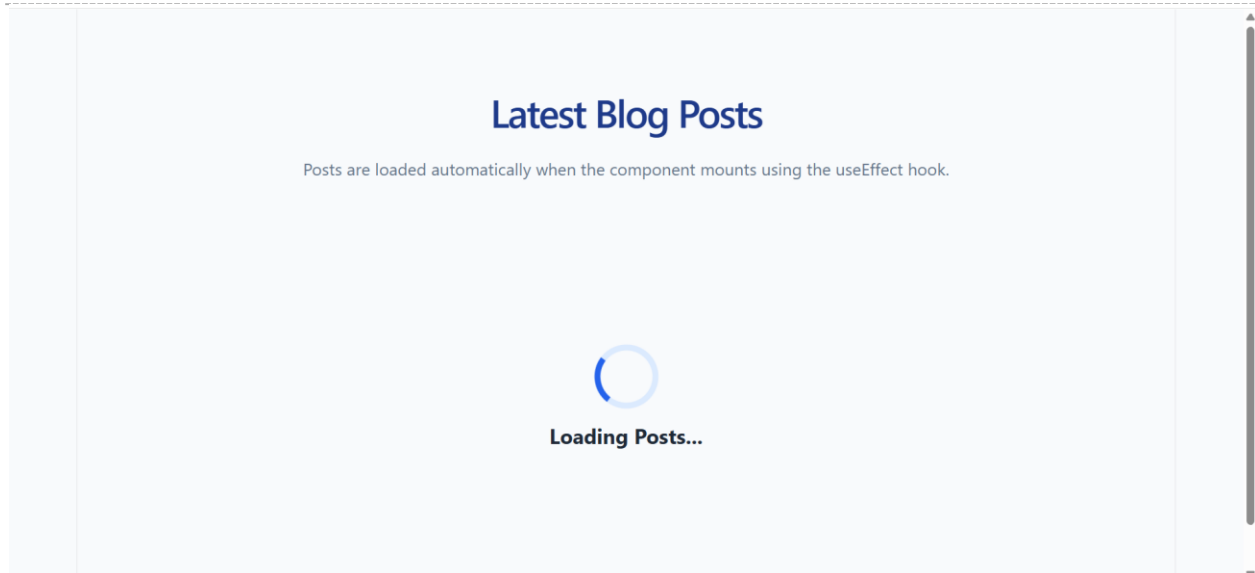
Successfully developed a React component that demonstrates the practical use of the component lifecycle through the useEffect hook. The project reinforced the understanding of executing side effects after component mounting, managing asynchronous API requests, and controlling component rendering based on application state. Additionally, implementing loading indicators and graceful error handling improved the overall user experience while strengthening knowledge of modern React development practices and asynchronous programming patterns.

Screenshots:



```
1 import { useEffect, useState } from "react";
2 import "./PostFetcher.css";
3
4 function PostFetcher() {
5   const [posts, setPosts] = useState([]);
6   const [loading, setLoading] = useState(true);
7   const [error, setError] = useState("");
8
9   useEffect(() => {
10     async function fetchPosts() {
11       try {
12         const response = await fetch("https://dummyjson.com/posts?limit=6");
13
14         if (!response.ok) {
15           throw new Error("Failed to load posts.");
16         }
17
18         const data = await response.json();
19         setPosts(data.posts);
20       } catch (err) {
21         setError(err.message);
22       } finally {
23         setLoading(false);
24       }
25     }
26
27     fetchPosts();
28   }, []);
29 }
```

Screenshots:



Latest Blog Posts

Posts are loaded automatically when the component mounts using the useEffect hook.

Article #1

His mother had always taught him

His mother had always taught him not to ever think of himself as better than others. He'd tried to live by this motto. He never looked down on those who were less fortunate or who had less money than him. But the stupidity of the group of people he was talking to made him change his mind.

Article #2

He was an expert but not in a discipline

He was an expert but not in a discipline that anyone could fully appreciate. He knew how to hold the cone just right so that the soft server ice-cream fell into it at the precise angle to form a perfect cone each and every time. It had taken years to perfect and he could now do it without even putting any

Article #3

Dave watched as the forest burned up on the hill.

Dave watched as the forest burned up on the hill, only a few miles from her house. The car had been hastily packed and Marta was inside trying to round up the last of the pets. Dave went through his mental list of the most important papers and documents that they couldn't leave behind. He scolded himself for not

Latest Blog Posts

Posts are loaded automatically when the component mounts using the useEffect hook.



Failed to fetch